ECON 8185-002 Homework 6

Sauhard Srivastava - sriva238@umn.edu

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The Krusell-Smith (1998) economy

In this homework I attempt to solve for the approximate equilibrium of an incomplete markets economy with aggregate productivity shocks as in Krussel-Smith (1998).

The economy consists of a unit mass of infinitely lived agents. There is only one good per period and agents can save in only one asset- capital. Each agent is endowed with one unit of time, which gives rise to $\epsilon\bar{l}$ units of labor input, where ϵ is stochastic and can take on the value zero or one. The agent's maximization problem is given by:

$$\max_{\{c_t, k_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma} \qquad \text{s.t.}$$
$$c_t + k_{t+1} = (1 + r_t - \delta)k_t + w_t \epsilon_t \bar{l}$$
$$c_t, k_{t+1} \geq 0$$

The production technology is CRS: $Y_t = z_t K_t^{\alpha} L_t^{1-\alpha}$ implying that:

$$w_t = (1-\alpha) z_t \left(\frac{K_t}{L_t}\right)^{\alpha}$$

$$R_t = (1 + r_t - \delta) = \alpha z_t \left(\frac{K_t}{L_t}\right)^{\alpha-1} + (1-\delta)$$

z_t is stochastic, follows a first order Markov structure, and can take on two values (z_g, z_b). The individual and aggregate shocks are correlated, and the individuals' shocks are assumed to satisfy a law of large numbers. Thus, the only exogenous source of aggregate uncertainty in the economy is the aggregate productivity shock. The number of agents who are unemployed always equals u_g in good times and u_b in bad times.

We will define a recursive equilibrium. However, unlike Aiyagari's (1994) economy, the agent's recursive problem must now include the aggregate state variables. The aggregate state is (Γ, z), where Γ denotes

the current distribution of consumers over holdings of capital and employment status. Therefore, in order to forecast prices tomorrow, the problem must also include laws of motion of the aggregate state variables. The law of motion for z is given by it's transition matrix. The law of motion for Γ is denoted H. The consumer's recursive problem is then:

$$V(k,\epsilon;\Gamma,z) = \max_{c,k'} \frac{c^{1-\gamma}}{1-\gamma} + \beta \sum_{z'} \pi(z'|z) \sum_{\epsilon'} \pi(\epsilon'|\epsilon,z) V(k',\epsilon';\Gamma',z')$$

$$\text{s.t.}$$

$$c + k' = R(\Gamma,z)k + w(\Gamma,z)\epsilon\bar{l}$$

$$\Gamma' = H(\Gamma,z,z')$$

$$c, k' \geq 0$$

Definition of the Equilibrium

A recursive competitive equilibrium is defined by:

- Pricing functions $R(\Gamma,z),\ w(\Gamma,z)$
- A value function $V(k,\epsilon;\Gamma,z)$
- Policy functions $k' = g(k,\epsilon;\Gamma,z),\ c = g_c(k,\epsilon;\Gamma,z)$

such that:

- Given prices, the policy functions solve the consumer's problem, and the value function attains the maximum.
- Given prices, the firm chooses K, L to maximize profits:

$$w(\Gamma,z) = (1-\alpha)z\left(\frac{K}{L}\right)^{\alpha}$$

$$R(\Gamma,z) = \alpha z\left(\frac{K}{L}\right)^{\alpha-1} + (1-\delta)$$

- The law of motion of Γ, H, is implied by the policy functions.
- Markets clear for all s:

$$K = \iint k d\Gamma$$

$$L = (1-u_s)\bar{l} \qquad \forall s$$

$$\iint g_c(k,\epsilon;\Gamma,z)d\Gamma + \iint g(k,\epsilon;\Gamma,z)d\Gamma = zK^{\alpha}L^{1-\alpha} + (1-\delta)K$$

Approximate Equilibrium

The endogenous aggregate state variable of the economy, Γ, is a high-dimensional object which makes this problem computationally difficult to solve. The Krusell-Smith way of dealing with this problem is to assume that agents are boundedly rational in their perceptions of how Γ evolves over time and solve for an approximate equilibrium using an approximate law of motion H_I that depends on only the first I moments of Γ. Krusell-Smith (1998) show that for this model, $I = 1$ i.e., using only the mean of the distribution as the aggregate state variable is sufficient to obtain an extremely good estimate of the equilibrium. They use a log-linear law of motion for the average/aggregate capital stock:

$$\ln K_{t+1} = a_0 + a_1 \ln K_t \qquad \text{for } z = z_b$$

$$\ln K_{t+1} = b_0 + b_1 \ln K_t \qquad \text{for } z = z_g$$

The consumer's recursive problem is then much simpler and computationally tractable:

$$V(k, \epsilon; K, z) = \max_{c,k'} \frac{c^{1-\gamma}}{1-\gamma} + \beta \sum_{z'} \pi(z'|z) \sum_{\epsilon'} \pi(\epsilon'|\epsilon, z) V(k', \epsilon'; K', z')$$

$$\text{s.t.-}$$

$$c + k' = R(K, z)k + w(K, z)\epsilon \bar{l}$$

$$\ln K' = a_0 + a_1 \ln K \qquad \text{for } z = z_b$$

$$\ln K' = b_0 + b_1 \ln K \qquad \text{for } z = z_g$$

$$c, k' \geq 0$$

The optimality conditions are then given by:

$$(R(K, z)k + w(K, z)\epsilon \bar{l} - k')^{-\gamma} \geq \beta \sum_{(\epsilon', z')} \pi(\epsilon', z'|\epsilon, z)[R'(K', z') \left(R'(K', z')k' + w'(K', z')\epsilon' \bar{l} - k''\right)^{-\gamma}]$$

$$\ln K' = a_0 + a_1 \ln K \qquad \text{for } z = z_b$$

$$\ln K' = b_0 + b_1 \ln K \qquad \text{for } z = z_g$$

$$w(K, z) = (1-\alpha) z \left(\frac{K}{L}\right)^{\alpha}$$

$$R(K, z) = \alpha z \left(\frac{K}{L}\right)^{\alpha-1} + (1-\delta)$$

$$L = (1 - u_s)\bar{l} \qquad \forall s$$

The KS algorithm

- Choose I and make an initial guess for H_I, in this case, $I = 1$ and H_I is chosen to have a log-linear form, i.e., guess a_0, a_1, b_0, b_1. Make an initial guess for Γ_0 and the corresponding K_0.

- Solve for the Consumer's saving policy function $g(k, \epsilon; K, z)$ using the guessed aggregate law of motion.
- Using only the consumer's decision rule, simulate a time series of savings by N agents for T periods, N, T large. Use the realized K_t from the simulation each period as the state variable for each agent's policy function.
- Use the stationary region of the simulated data to estimate a set of parameters for the functional forms assumed above. At this stage, we obtain a measure of goodness of fit.
- If the estimation gives parameter values that are very close to those guessed initially and the goodness of fit is satisfactory, stop. If the parameter values have converged but the goodness of fit is not satisfactory, increase I or, try a different functional form for H.

Solving for an approximate equilibrium

The model parameterization closely follows Krusell-Smith (1998):

Parameter	Value
β	0.99
γ	1
α	0.36
δ	0.025
z_g	1.01
z_b	0.99
u_g	0.1
u_b	0.04
$\bar{l}$	0.309

Let $\pi_{ss'\epsilon\epsilon'}$ denote the transition probability from (z, ϵ) to (z', ϵ'). The Markov matrix can be expressed as:

$$\Pi = \begin{bmatrix} \pi_{bb00} & \pi_{bb01} & \pi_{bg00} & \pi_{bg01} \\ \pi_{bb10} & \pi_{bb11} & \pi_{bg10} & \pi_{bg11} \\ \pi_{gb00} & \pi_{gb01} & \pi_{gg00} & \pi_{gg01} \\ \pi_{gb10} & \pi_{gb11} & \pi_{gg10} & \pi_{gg11} \end{bmatrix}$$

Let the transition matrix of the aggregate shock be given by:

$$\Pi_z = \begin{bmatrix} \pi_{bb} & \pi_{bg} \\ \pi_{gb} & \pi_{gg} \end{bmatrix} = \begin{bmatrix} \frac{7}{8} & \frac{1}{8} \\ \frac{1}{8} & \frac{7}{8} \end{bmatrix}$$

Let:

- $\pi_{00|bb} = 0.6$, then $\pi_{01|bb} = 0.4$
- $\pi_{00|gg} = 1/3$, then $\pi_{01|gg} = 2/3$
- $\pi_{00|gb} = 1.25\pi_{00|bb}$
- $\pi_{00|bg} = 0.75\pi_{00|gg}$

These assumptions together with Π_z can be used to obtain the first and the third rows of Π. For the second and the fourth rows use the following condition:

$$u_s\pi_{00|ss'} + (1 - u_s)\pi_{01|ss'} = u'_s \qquad \forall s, s'$$

where $\pi_{xy|ss'} = \dfrac{\pi_{ss'zw}}{\pi_{ss'}}$ $\forall s, s'$. Using this parameterization, the transition matrix is given as:

$$\Pi = \begin{bmatrix} 0.5250 & 0.3500 & 0.0313 & 0.0938 \\ 0.0389 & 0.8361 & 0.0021 & 0.1229 \\ 0.0938 & 0.0313 & 0.2917 & 0.5833 \\ 0.0091 & 0.1159 & 0.0243 & 0.8507 \end{bmatrix}$$

For the sake of comparison, the coefficients obtained by Krusell-Smith (1998) are:

$$\ln K' = 0.085 + 0.965 \ln K \qquad R^2 = 0.999998 \qquad \text{for } z = z_b$$

$$\ln K' = 0.095 + 0.962 \ln K \qquad R^2 = 0.999998 \qquad \text{for } z = z_g$$

Results

I solve the model using MATLAB® R2022b. I use the endogenous grid method for solving the policy functions by iterating on the Euler equation. I use MATLAB's `griddedInterpolant()` for spline interpolation of the policy function in the k and K dimensions. Following, Krusell-Smith, I set $N = 5000$ and $T = 11000$. Note that the laws of motion obtained by Krusell-Smith imply that the ergodic set of K is given by $[11.34266, 12.18249]$. In order to obtain coefficients close to their solution, I scaled $\bar{l}$ so that the steady state is in this ergodic set.[1]

I started the iterations of the KS algorithm using the coefficients obtained by Krusell-Smith as the initial guess. However, this was not necessary for obtaining the solution, but helpful in saving time. Table 1 shows the results I obtained. Figure 1 plots the policy function. Figure 2 shows the scatter plot of today's and tomorrow's aggregate capital stock for each state generated by the sample. Figure 3 show's the fundamental accuracy plot as recommended by den Haan (2010). Figure 4 shows the stationary

[1] This was not necessary, alternative scaling is to set $\bar{l} = 1/u_b$ so that $L_b = 1$. This is done in Maliar, Maliar and Vali (2010). Using this scaling my results were close to those obtained by them.

marginal density of wealth. Figure 5 shows the Lorenz curve for wealth. Table 2 shows the distribution statistics for wealth, consumption and income including the Gini coefficients.

Discussion

- Starting from the specified initial guess of Krusell-Smith's original coefficients, and a given tolerance, the algorithm took 30 iterations to converge in about 11 minutes on an 11th gen Intel Core-i7 quad core PC (see published code).
- The coefficients in Table 1 are very close to those obtained by Krusell-Smith (1998) They are robust to the initial guess chosen for the KS algorithm.
- Unlike Krusell-Smith (1998) results, R^2 is unable to achieve a five digit 0.99999 accuracy, while the coefficients converged. As the KS algorithm suggests, there might be a need to include higher moments to get better accuracy.
- The fundamental accuracy plot suggests that the perceived and the actually generated aggregate laws of motion are fairly close to each other. However, on closer examination, small deviations can be seen. We cannot completely conclude that the results are accurate. However, including the variance of the distribution as a state variable may improve the fit.
- Gini coefficient for wealth is 0.2440, which is very close to what Krusell-Smith (1998) obtained. Income inequality is marginally less that the wealth inequality whereas consumption inequality is significantly lesser highlighting the risk sharing and consumption smoothing going on in the background.

Table 1: Coefficients of the aggregate law of motion

	const.	**coeff.**	R^2
$z = z_b$	0.08252896	0.96447503	0.99986777
$z = z_g$	0.09158663	0.96292958	0.99993991

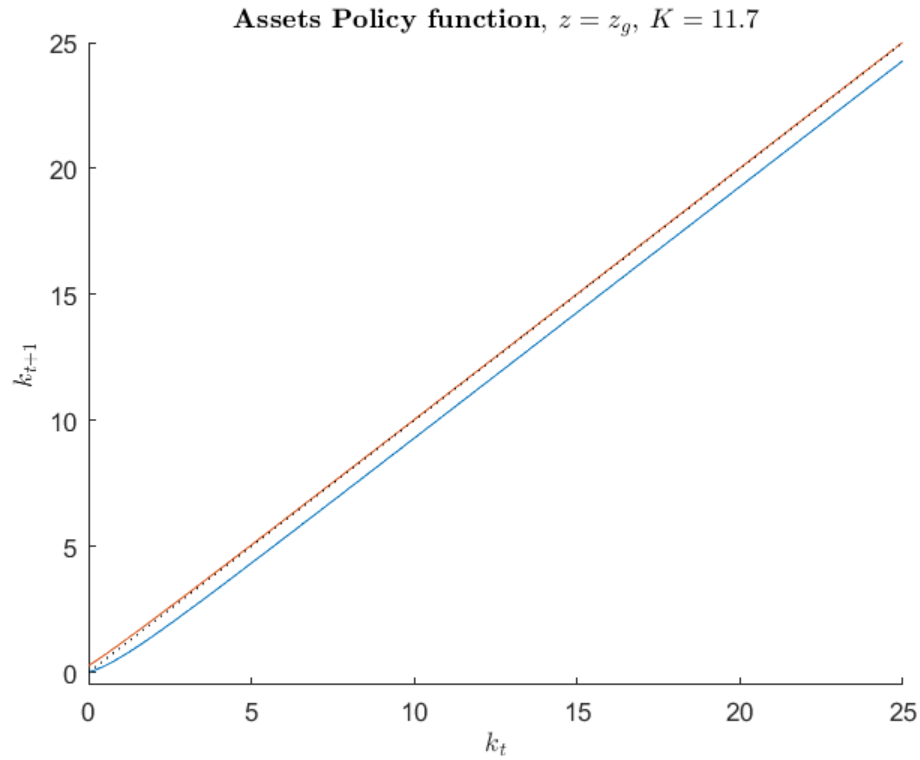

Figure 1: Assets Policy function

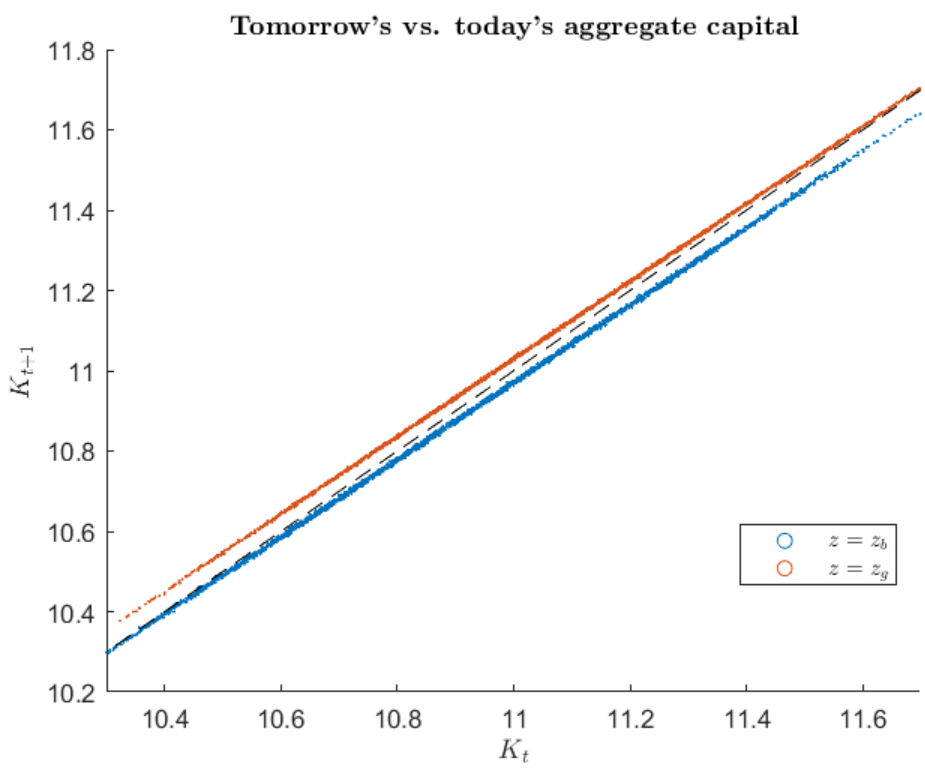

Figure 2: Tomorrow's vs today's aggregate capital

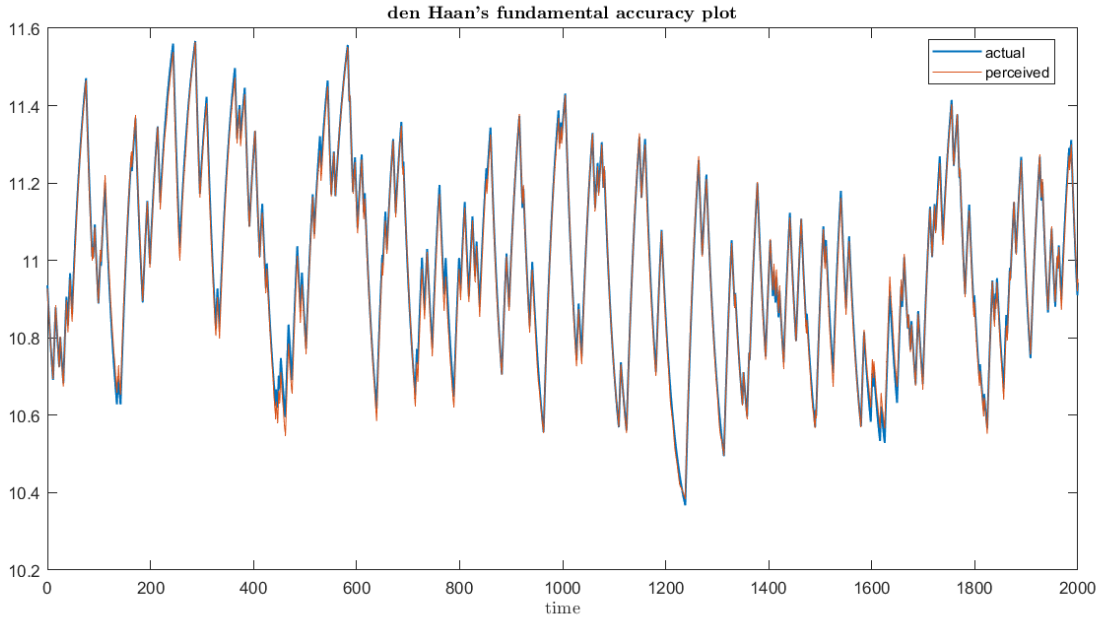

Figure 3: den Haan's fundamental accuracy plot

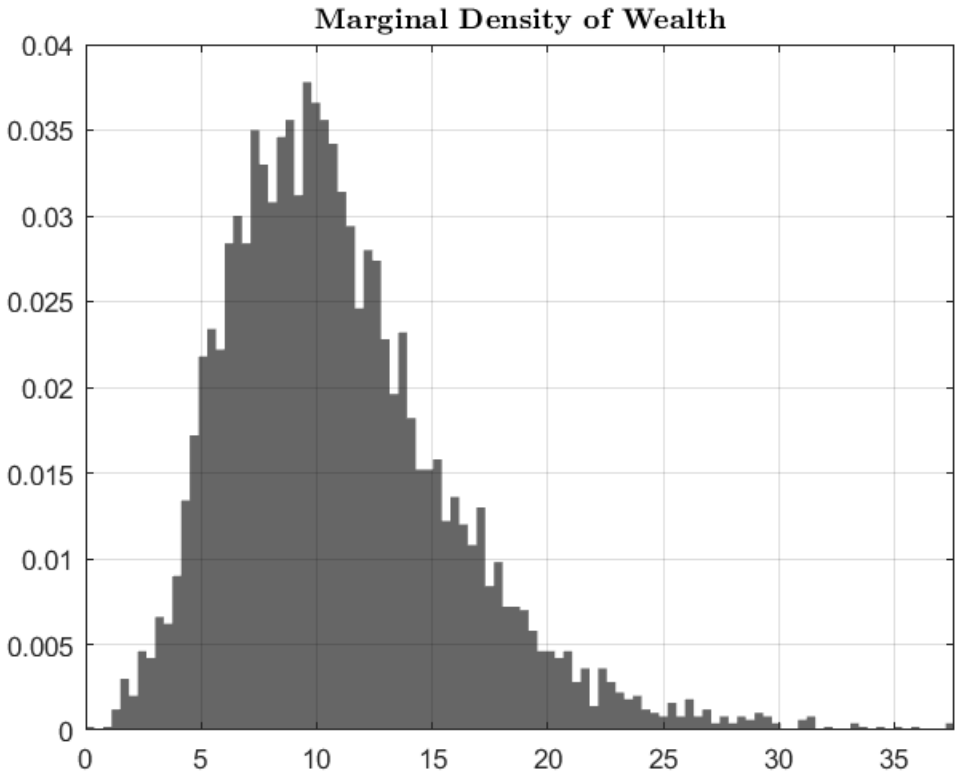

Figure 4: Stationary marginal density of wealth

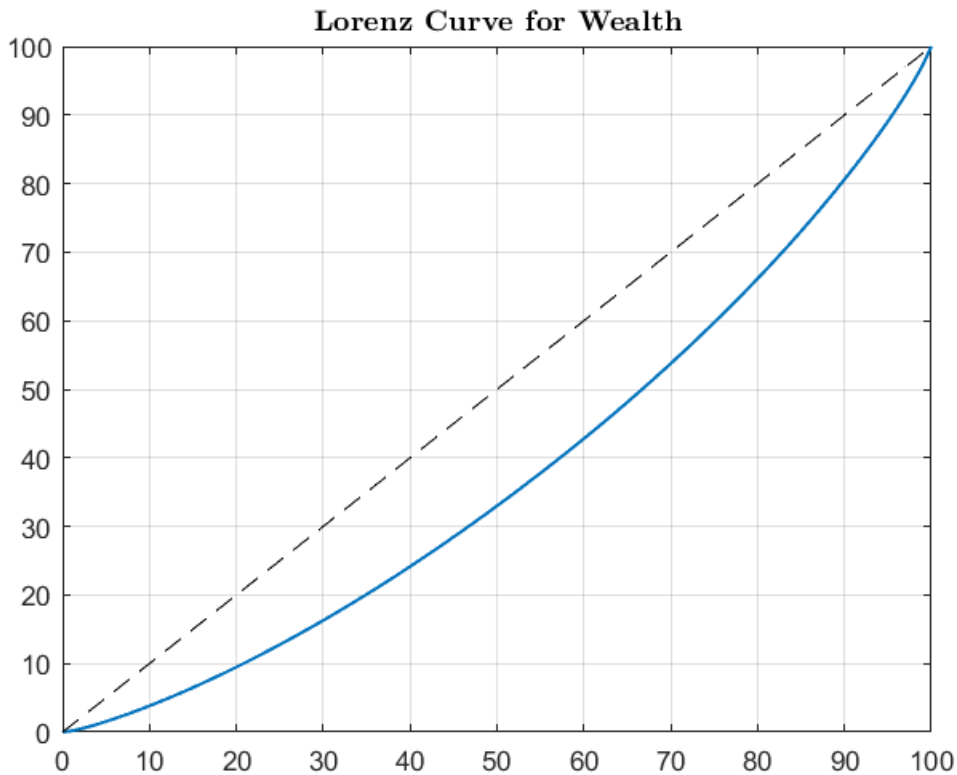

Figure 5: Lorenz curve for wealth

Table 2: Distribution statistics

	Top 1%	Top 5%	Top 10%	Top 20%	Top 30%	Gini coeff.
wealth	2.73	10.92	19.39	33.86	46.24	0.24
income	2.63	10.58	18.85	33.08	45.34	0.23
consumption	1.53	6.95	12.62	23.35	33.72	0.06

Code Readme

All code is created and executed with MATLAB® R2022b and may not run properly on older versions of MATLAB®. Please also refer to published code (pdf) files to see the output. The code files are as follows:

File	Description
`kssolver.m`	This script runs the KS algorithm and obtains the coefficients of the ALM. It also plots the figures.
`pisolver.m`	This script solves for Π matrix using the method described above.
`kspolicy.m`	This function solves for the policy function of assets using the endogenous grid method.
`shocksim.m`	This function simulates the aggregate and idiosyncratic shocks.